

ECE-3103

Instruction Format in 8086

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Introduction

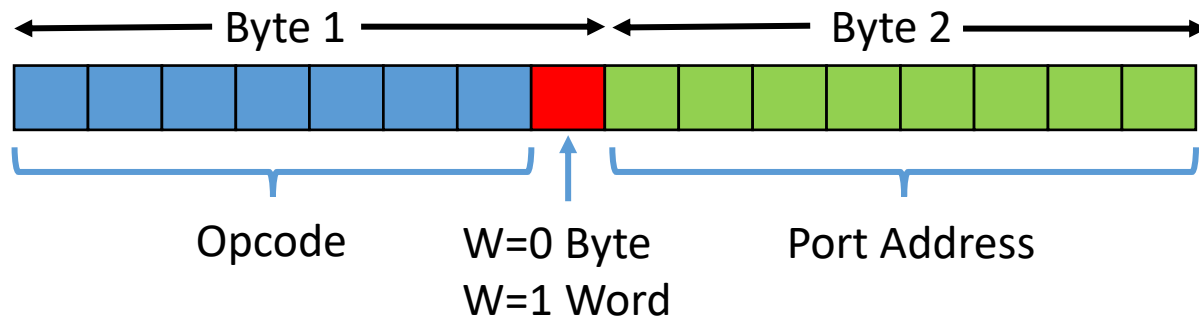
- Suppose there is an instruction, such as: MOV CX, source.
- There are 32 ways to specify the source of the operand in this instruction.
- The source of the operand can be any one of eight 16-bit registers(AX, BX, CX, DX, SP, DI, BP, SI), or a memory location specified by any one of 24 memory addressing modes.
- Each of the 32 possible instructions requires a different binary code.
- If CX is made the source rather than the destination, then there are 32 ways of specifying the destinations.
- Again Each of these 32 possible instructions requires a different binary code.

Introduction

- There are thus 64 different codes for MOV instructions using CX as a source or as a destination.
- Likewise, another 64 codes are required to specify all the possible MOVs using CL as a source or destination, and 64 more are required to specify all the possible MOVs using CH as a source or a destination.
- So this is problematic.
- So we use a template for each basic instruction type and fill in bits within this template to indicate the desired addressing mode, data type, etc.
- In other words, we build up the instruction codes on a bit-by-bit basis.

Instruction Coding Format Example

- Let the instruction is IN AL, 05H.
- Coding template for this instruction is given below:



- Opcode for IN 1110010. Here as the above instruction input a byte from an 8-bit port to AL, so W=0. Here 8-bit port address of 05H is 00000101 in binary number.
- So the binary code for IN AL, 05H is:



- And corresponding Hex code is E4H, 05H.

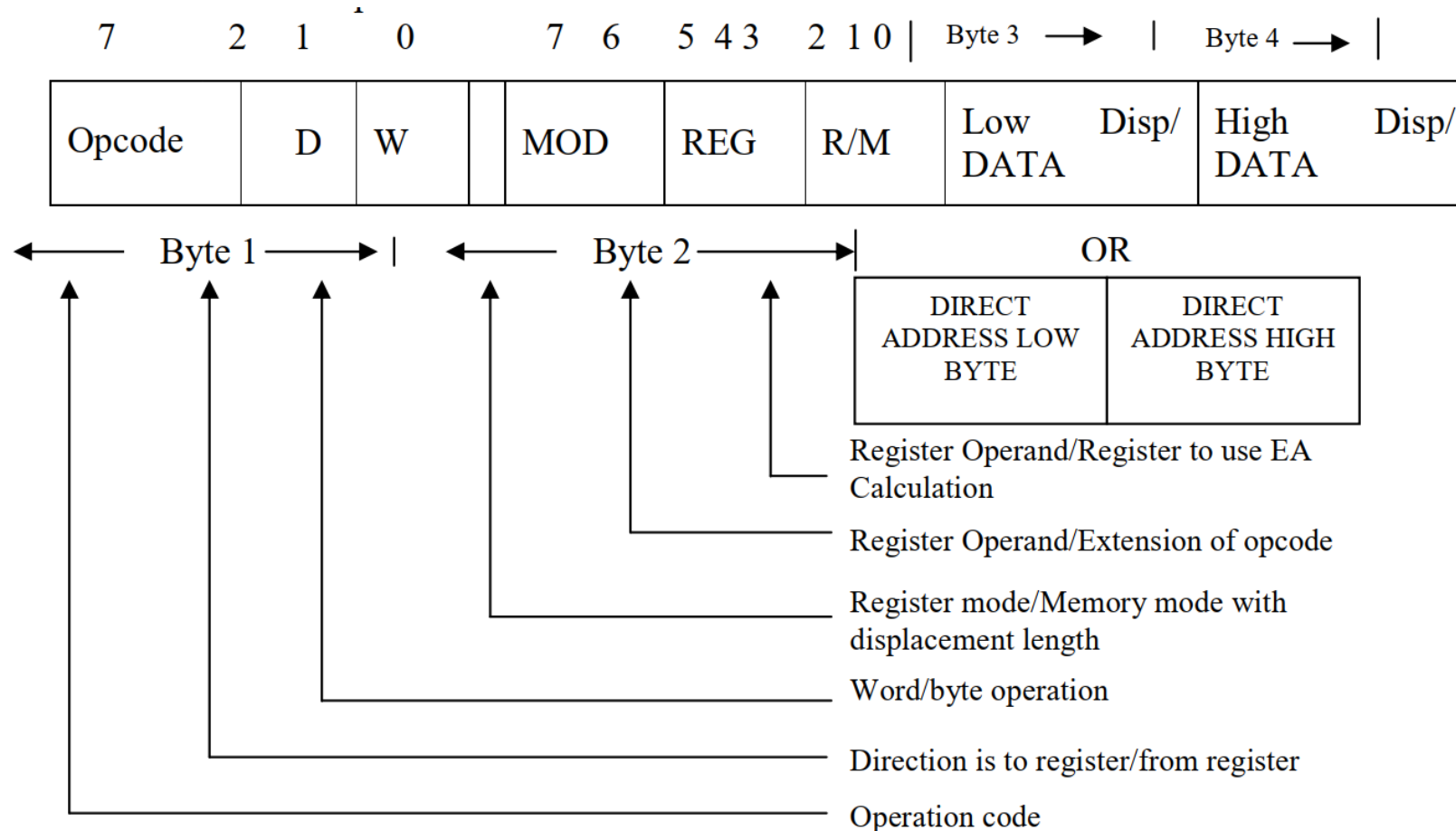
Instruction Coding Format Example

- Now, for instruction IN AX, 2305H, binary coding will be :



- Here opcode is 1110010.
- Instruction inputs a word from an 16-bit port to AX, So W=1.
- Again as lower byte address is 05 we take binary 00000101 as 2nd byte.
- As higher byte address is 23 we take binary 00100011 as 3rd byte.
- So the corresponding Hex code is: E5H, 05H, 23H.

MOV Instruction Coding Format



❑ The OP code field occupies 6-bits. It defines the operation to be carried out by the instruction.

MOV Instruction Coding Format

- Register Direct bit (D) occupies one bit. It defines whether the register operand in byte 2 is the source or destination operand.
 - ✓ D=1 Specifies that the register operand is the destination operand.
 - ✓ D=0 indicates that the register is a source operand.
- Data size bit (W) defines whether the operation to be performed is an 8 bit or 16 bit data.
 - ✓ W=0 indicates 8 bit operation.
 - ✓ W=1 indicates 16 bit operation.

MOV Instruction Coding Format

- The second byte of the instruction usually identifies whether one of the operands is in memory or whether both are registers.
- This byte contains 3 fields. These are the mode (MOD) field, the register (REG) field and the Register/Memory (R/M) field.

MOD (2 bits)	Interpretation
00	Memory mode with no displacement follows except for 16 bit displacement when R/M=110
01	Memory mode with 8 bit displacement
10	Memory mode with 16 bit displacement
11	Register mode (no displacement)

MOV Instruction Coding Format

- Register field occupies 3 bits. It defines the register for the first operand which is specified as source or destination by the D bit.

REG	W=0	W=1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

MOV Instruction Coding Format

- ❑ The R/M field occupies 3 bits. The R/M field along with the MOD field defines the second operand

For MOD=11

MOD / R/M	Memory Mode (EA Calculation)			Register Mode	
	00	01	10	W=0	W=1
000	(BX)+(SI)	(BX)+(SI)+d8	(BX)+(SI)+d16	AL	AX
001	(BX) + (DI)	(BX)+(DI)+d8	(BX)+(DI)+d16	CL	CX
010	(BP)+(SI)	(BP)+(SI)+d8	(BP)+(SI)+d16	DL	DX
011	(BP)+(DI)	(BP)+(DI)+d8	(BP)+(DI)+d16	BL	BX
100	(SI)	(SI) + d8	(SI) + d16	AH	SP
101	(DI)	(DI) + d8	(DI) + d16	CH	BP
110	d16	(BP) + d8	(BP) + d16	DH	SI
111	(BX)	(BX) + d8	(BX) + d16	BH	DI

MOV Instruction Coding Format

- For other MODs except 11, register is considered as the first operand.
- In MOD=11, as both the operands are register, we have to consider any one of them as the first operand, and another as the second operand.
- In every cases first operand determines the REG number and second operand determines the R/M number.
- The opcode for MOV instruction is 100010.

MOV Instruction Coding Examples

❑ MOV SP, BX:

- If we take SP as the first operand, then as SP is the destination, D=1; as it moves a word(16-bit), W=1.
- As both the operand are registers, MOD=11.

MOD (2 bits)	Interpretation
00	Memory mode with no displacement follows except for 16 bit displacement when R/M=110
01	Memory mode with 8 bit displacement
10	Memory mode with 16 bit displacement
11	Register mode (no displacement)

MOV Instruction Coding Examples

❑ MOV SP, BX:

- As SP is taken as the first operand, we will get REG=100.

REG	W=0	W=1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

MOV Instruction Coding Examples

❑ MOV SP, BX:

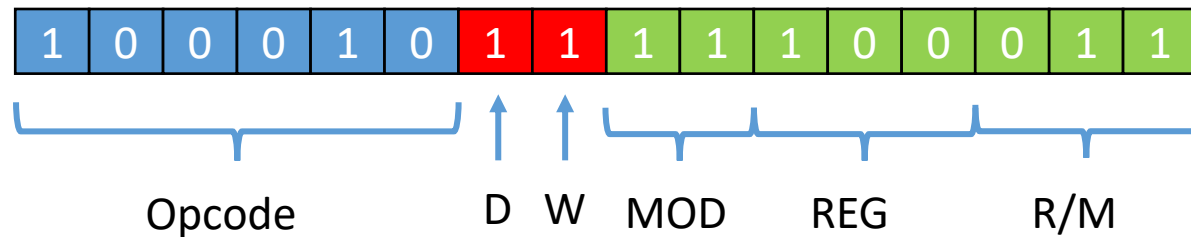
- As BX is taken as the second operand, we will get R/M=011.

MOD / R/M	Memory Mode (EA Calculation)			Register Mode	
	00	01	10	W=0	W=1
000	(BX)+(SI)	(BX)+(SI)+d8	(BX)+(SI)+d16	AL	AX
001	(BX) + (DI)	(BX)+(DI)+d8	(BX)+(DI)+d16	CL	CX
010	(BP)+(SI)	(BP)+(SI)+d8	(BP)+(SI)+d16	DL	DX
011	(BP)+(DI)	(BP)+(DI)+d8	(BP)+(DI)+d16	BL	BX
100	(SI)	(SI) + d8	(SI) + d16	AH	SP
101	(DI)	(DI) + d8	(DI) + d16	CH	BP
110	d16	(BP) + d8	(BP) + d16	DH	SI
111	(BX)	(BX) + d8	(BX) + d16	BH	DI

MOV Instruction Coding Examples

❑ MOV SP, BX:

- So the coding format will be (taking SP as first operand).



- Corresponding Hex code is: 8BH, E3H.
- What happens if we take BX as first operand?

MOV Instruction Coding Examples

❑ MOV SP, BX:

- If we take BX as the first operand, then as BX is the source, D=0; as it moves a word(16-bit), W=1.
- As both the operand are registers, MOD=11.

MOD (2 bits)	Interpretation
00	Memory mode with no displacement follows except for 16 bit displacement when R/M=110
01	Memory mode with 8 bit displacement
10	Memory mode with 16 bit displacement
11	Register mode (no displacement)

MOV Instruction Coding Examples

❑ MOV SP, BX:

- As BX is taken as the first operand, we will get REG=011.

REG	W=0	W=1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

MOV Instruction Coding Examples

❑ MOV SP, BX:

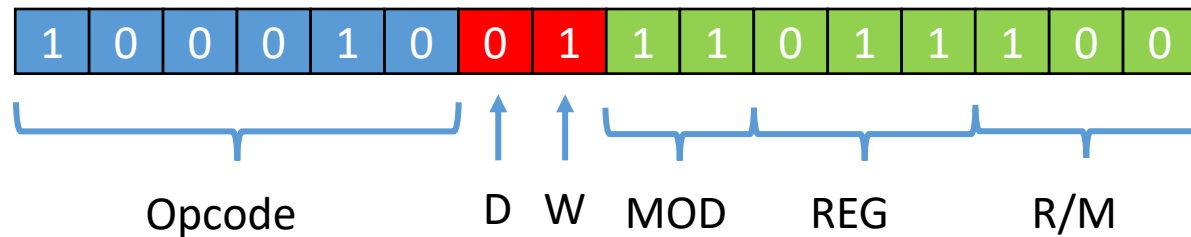
- As SP is taken as the second operand, we will get R/M=100.

MOD / R/M	Memory Mode (EA Calculation)			Register Mode	
	00	01	10	W=0	W=1
000	(BX)+(SI)	(BX)+(SI)+d8	(BX)+(SI)+d16	AL	AX
001	(BX) + (DI)	(BX)+(DI)+d8	(BX)+(DI)+d16	CL	CX
010	(BP)+(SI)	(BP)+(SI)+d8	(BP)+(SI)+d16	DL	DX
011	(BP)+(DI)	(BP)+(DI)+d8	(BP)+(DI)+d16	BL	BX
100	(SI)	(SI) + d8	(SI) + d16	AH	SP
101	(DI)	(DI) + d8	(DI) + d16	CH	BP
110	d16	(BP) + d8	(BP) + d16	DH	SI
111	(BX)	(BX) + d8	(BX) + d16	BH	DI

MOV Instruction Coding Examples

❑ MOV SP, BX:

- So the coding format will be (taking BX as first operand).



- Corresponding Hex code is: 89H, DCH.
- Generated Hex code in both case is correct.

MOV Instruction Coding Examples

❑ MOV 43H[SI], DH:

- As DH is the register, we take DH as the first operand, then as DH is the source, D=0; as it moves a byte(8-bit), W=0.
- As one operand is memory with 8-bit displacement, MOD=01.

MOD (2 bits)	Interpretation
00	Memory mode with no displacement follows except for 16 bit displacement when R/M=110
01	Memory mode with 8 bit displacement
10	Memory mode with 16 bit displacement
11	Register mode (no displacement)

MOV Instruction Coding Examples

❑ MOV 43H[SI], DH:

- As DH is taken as the first operand, we will get REG=110.

REG	W=0	W=1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

MOV Instruction Coding Examples

❑ MOV 43H[SI], DH:

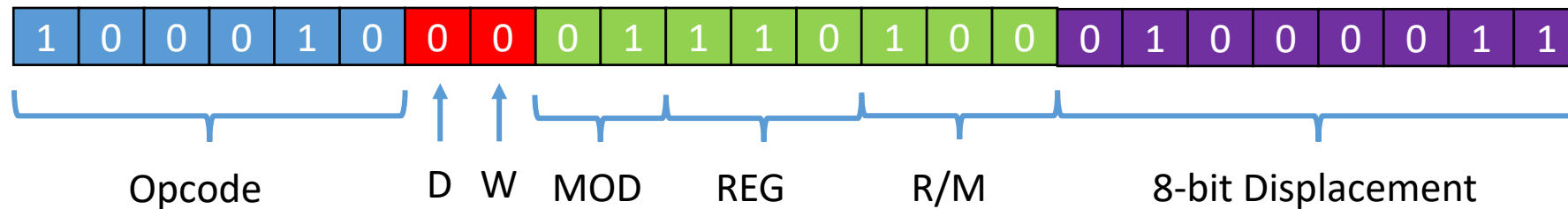
- As 43H[SI] is taken as the second operand, we will get R/M=100.

MOD / R/M	Memory Mode (EA Calculation)			Register Mode	
	00	01	10	W=0	W=1
000	(BX)+(SI)	(BX)+(SI)+d8	(BX)+(SI)+d16	AL	AX
001	(BX) + (DI)	(BX)+(DI)+d8	(BX)+(DI)+d16	CL	CX
010	(BP)+(SI)	(BP)+(SI)+d8	(BP)+(SI)+d16	DL	DX
011	(BP)+(DI)	(BP)+(DI)+d8	(BP)+(DI)+d16	BL	BX
100	(SI)	(SI) + d8	(SI) + d16	AH	SP
101	(DI)	(DI) + d8	(DI) + d16	CH	BP
110	d16	(BP) + d8	(BP) + d16	DH	SI
111	(BX)	(BX) + d8	(BX) + d16	BH	DI

MOV Instruction Coding Examples

❑ MOV 43H[SI], DH:

- So the coding format will be (taking DH as first operand and 43H as the 8-bit displacement).



- Corresponding Hex code is: 88H, 74H, 43H.
- Here 3rd byte denotes the displacement.

MOV Instruction Coding Examples

□ MOV CL, [BX]:

- As CL is the register, we take CL as the first operand, then as CL is the destination, D=1; as it moves a byte(8-bit), W=0.
- As one operand is memory with no displacement, MOD=00.

MOD (2 bits)	Interpretation
00	Memory mode with no displacement follows except for 16 bit displacement when R/M=110
01	Memory mode with 8 bit displacement
10	Memory mode with 16 bit displacement
11	Register mode (no displacement)

MOV Instruction Coding Examples

❑ **MOV CL, [BX]:**

- As CL is taken as the first operand, we will get REG=001.

REG	W=0	W=1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

MOV Instruction Coding Examples

❑ MOV CL, [BX]:

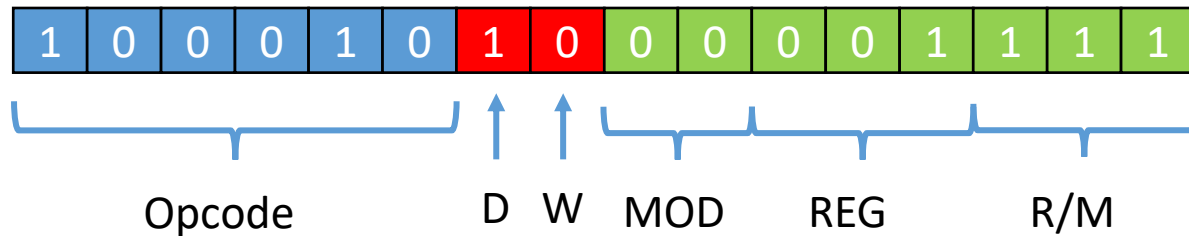
- As [BX] is taken as the second operand, we will get R/M=111.

MOD / R/M	Memory Mode (EA Calculation)			Register Mode	
	00	01	10	W=0	W=1
000	(BX)+(SI)	(BX)+(SI)+d8	(BX)+(SI)+d16	AL	AX
001	(BX) + (DI)	(BX)+(DI)+d8	(BX)+(DI)+d16	CL	CX
010	(BP)+(SI)	(BP)+(SI)+d8	(BP)+(SI)+d16	DL	DX
011	(BP)+(DI)	(BP)+(DI)+d8	(BP)+(DI)+d16	BL	BX
100	(SI)	(SI) + d8	(SI) + d16	AH	SP
101	(DI)	(DI) + d8	(DI) + d16	CH	BP
110	d16	(BP) + d8	(BP) + d16	DH	SI
111	(BX)	(BX) + d8	(BX) + d16	BH	DI

MOV Instruction Coding Examples

❑ MOV CL, [BX]:

- So the coding format will be (taking CL as first operand and no displacement).

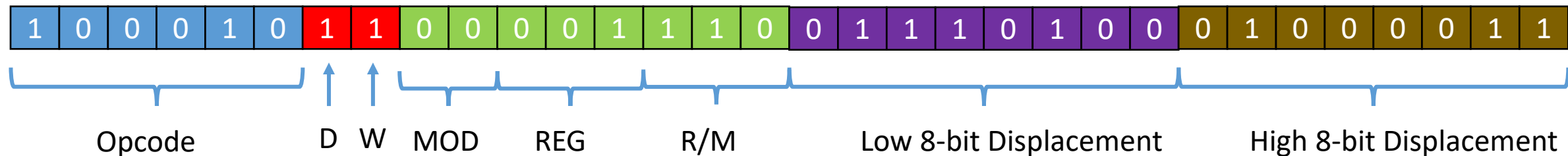


- Corresponding Hex code is: 8AH, 0FH.

MOV Instruction Coding Examples

❑ MOV CX, [437AH]:

- Do it by yourself.
- Coding format will be (taking CX as the first operand, 74H as the low displacement and 43H as the high displacement) :



- Corresponding Hex code is: 8BH, 0EH, 74H, 43H.

MOV Instruction Coding Examples

□ **MOV CS: [BX], DL:**

- This instruction copies a byte from the DL register to a memory location whose effective address is contained in the BX register.
- Normally the effective address in BX will be added to the data segment base in DS to produce the physical memory address.
- In this instruction, CS: in front of [BX] indicates that we want the BIU to add the effective address to the code segment base in the CS to produce the physical address.
- The CS: is called **segment over prefix (SOP)**.

MOV Instruction Coding Examples

❑ **MOV CS: [BX], DL:**

- When an instruction containing SOP is coded a 8-bit code for the SOP is put in memory before the code for the rest of the instruction.
- The code byte for the SOP has the format 001XX110

xx	Segment register
00	ES
01	CS
10	SS
11	DS

MOV Instruction Coding Examples

❑ **MOV CS: [BX], DL:**

- Here the segment override prefix (SOP) byte for CS is 01.

xx	Segment register
00	ES
01	CS
10	SS
11	DS

- So the first byte is 00101110.

MOV Instruction Coding Examples

❑ MOV CS: [BX], DL:

- As DL is the register, we take DL as the first operand, then as DL is the source, D=0; as it moves a byte(8-bit), W=0.
- As one operand is memory with no displacement, MOD=00.

MOD (2 bits)	Interpretation
00	Memory mode with no displacement follows except for 16 bit displacement when R/M=110
01	Memory mode with 8 bit displacement
10	Memory mode with 16 bit displacement
11	Register mode (no displacement)

MOV Instruction Coding Examples

❑ MOV CS: [BX], DL:

- As DL is taken as the first operand, we will get REG=010.

REG	W=0	W=1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

MOV Instruction Coding Examples

❑ MOV CS: [BX], DL:

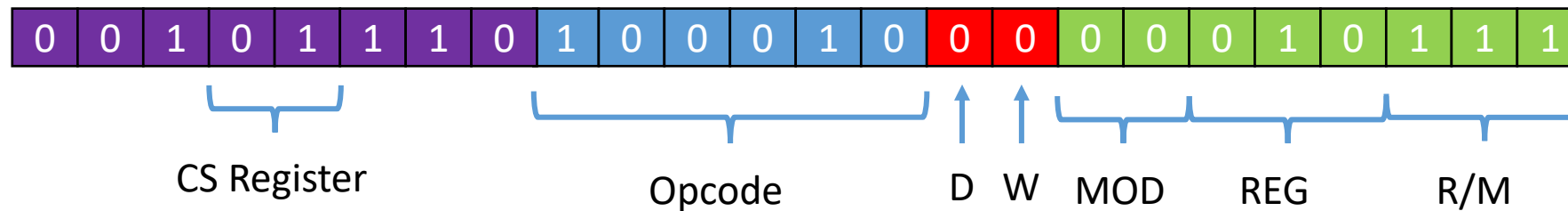
- As [BX] is taken as the second operand, we will get R/M=111.

MOD / R/M	Memory Mode (EA Calculation)			Register Mode	
	00	01	10	W=0	W=1
000	(BX)+(SI)	(BX)+(SI)+d8	(BX)+(SI)+d16	AL	AX
001	(BX) + (DI)	(BX)+(DI)+d8	(BX)+(DI)+d16	CL	CX
010	(BP)+(SI)	(BP)+(SI)+d8	(BP)+(SI)+d16	DL	DX
011	(BP)+(DI)	(BP)+(DI)+d8	(BP)+(DI)+d16	BL	BX
100	(SI)	(SI) + d8	(SI) + d16	AH	SP
101	(DI)	(DI) + d8	(DI) + d16	CH	BP
110	d16	(BP) + d8	(BP) + d16	DH	SI
111	(BX)	(BX) + d8	(BX) + d16	BH	DI

MOV Instruction Coding Examples

❑ MOV CS: [BX], DL:

- So the coding format will be (taking DL as first operand and no displacement).



- Corresponding Hex code is: 2EH, 88H, 17H.